

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 15

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 3 from:</i> • Disposal of items containing mercury: batteries, thermometers, fluorescent lights • Chemical plants producing chlorine using mercury electrodes • Combustion of coal • Gold mining (amalgamation)	3
01.2	<i>Any 3 from:</i> • Replacement of mercury thermometers with alcohol / electronic thermometers (discontinued uses) • Removal from effluents by reverse osmosis / activated carbon filters / ion exchange filters • Disposal / storage of wastes at high pH to reduce solubility • Encapsulation of mercury-containing hazardous waste in cement	3
01.3	• Most heavy metals are more soluble at low pH / less soluble at high pH • So at high pH they remain insoluble / immobile and cannot be leached into water or absorbed by organisms	2
01.4	• The effects of two pollutants interact to produce a different (greater) effect than either alone • Cadmium and zinc; ozone and sulfur dioxide	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-278; AQA spec 3.4.3.2.11.	
02.1	• As the (linear) purity / grade of a mineral deposit decreases, the amount of the mineral present increases exponentially / logarithmically	1
02.2	• $2400 / 40 = 60$ times	1

Question	Marking guidance	Marks
02.3	• Reserves: the amount that can be exploited now, economically, with existing technology (deposits above the cut-off ore grade) • Resource: all the material that could be exploited technically and economically now or in the future (with realistic price rises or new technology), including deep, low-grade or protected deposits • The resource is finite but the reserves change: they rise if prices rise or technology improves, and fall if prices fall	3
02.4	• Lowering the cut-off ore grade includes deposits of lower purity in the reserves • Because the amount of material rises exponentially as purity falls, a small drop in COOG adds a very large mass (eg 0.5-1.0% has 4x the copper of 1.0-2.0%)	2
02.5	• The total quantity (stock / resource base) is far greater than could ever be used • But most is at purities too low or in locations too inaccessible for current economic exploitation • The challenge is developing technologies (low-energy extraction such as bioleaching, phytomining; better exploration) that make low-grade deposits exploitable; energy needs rise rapidly as ore grade falls	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 135-136; AQA spec 3.2.3.3.	
03.1	• Soil loss increases as vegetation cover decreases: forest lowest (0.5), bare soil highest ($60 \text{ t ha}^{-1} \text{ yr}^{-1}$) - over 100 times more • Ploughing downslope loses nearly three times as much as contour ploughing for the same crop	2
03.2	<i>Any 4 from:</i> • Leaves / leaf litter intercept raindrops so soil particles are not dislodged by rain splash • Roots bind soil particles together • Vegetation acts as a windbreak reducing wind velocity and the kinetic energy available to carry particles • Plants increase infiltration (root channels, organic matter) so less surface runoff • Organic matter / humus from vegetation binds particles into stable peds	4
03.3	• Furrows running downslope act as channels / gulleys that increase the velocity (kinetic energy) of runoff • Furrows along the contour trap the water so it loses energy, deposits particles and infiltrates	2
03.4	• In the dry season the soil (especially low-clay soils) is loose and dry with little cohesion so wind blows it away • In the wet season heavy rain causes rain splash and surface runoff when infiltration capacity is exceeded	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.	
04.1	• Oxygen	1
04.2	• A change to one process causes other processes to alter as a direct result (chains of linked processes / feedback) • There is still a lot that is not understood about how atmospheric processes work	2

Question	Marking guidance	Marks
04.3	- The surface warmed in the tropics heats the air above; warm winds (south-westerlies from the Caribbean) carry heat energy to higher latitudes - Winds carry water vapour (evaporated from warm oceans) to areas that would otherwise receive little precipitation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-88; AQA spec 3.2.1.1.	
05.1	• 7×10^{11} tonnes	1
05.2	10% of 10 000 = 1000 billion tonnes 1000 / 700 x 100 = 143% (accept 142.9)	2
05.3	- Carbon is transferred into and out of rocks extremely slowly (fossilisation / sedimentation, weathering, volcanic release) - Residence time is millions of years so the flux to the atmosphere is tiny compared with respiration / photosynthesis / combustion	2
05.4	- CO₂ dissolves in the sea (forming carbonic acid / hydrogen carbonate ions) and is absorbed by phytoplankton through photosynthesis - Organisms such as molluscs, corals and planktonic organisms build calcium carbonate shells / exoskeletons - When they die the shells sink and sediment on the seabed, becoming limestone / chalk over long periods	3
05.5	- Aerobic respiration uses oxygen to break down organic compounds completely to CO₂ (and water), releasing much more energy - Without oxygen, anaerobic organisms only partly break down organic compounds, releasing carbon as methane (CH₄), which is later oxidised to CO₂ in the atmosphere	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-149; AQA spec 3.2.4.2.	
06.1	- Manufacture and use of CFCs and other ozone-depleting substances (ODSs) phased out then banned - HCFCs to be phased out (by 2030); some essential uses (eg halon fire extinguishers in aircraft) still permitted - A fund to help (developing) countries implement the Protocol; legally binding targets (unlike the 1985 Vienna Convention)	3
06.2	- Aerosols: pump-action sprays / roll-on deodorants (alternative process), or hydrocarbon propellants (propane, butane) - flammable / fire risk - Refrigerants: HCFCs (less stable so break down in the troposphere, but still some ozone depletion) then HFCs (no chlorine) - more expensive, less effective, powerful greenhouse gases	4
06.3	- CFCs are drained from the appliance and incinerated / broken down at high temperature - Products are CO₂ and acidic gases (HCl, HF) which are neutralised with crushed lime	2
06.4	CFCs are persistent, so those already released continue to reach the stratosphere and destroy ozone for many years; some ODSs (eg dichloromethane) are not banned	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 112-113; AQA spec 3.2.1.3.	

Question	Marking guidance	Marks
07.1	• A turbofan at the front forces bypass air around the inner cowling / core jet • The bypass air smooths the flow of the turbulent hot exhaust gases, reducing the turbulence that creates noise • The higher the ratio of bypass air to jet exhaust the quieter the engine (Trent 1000 quietest, Olympus noisiest)	3
07.2	<i>Any 2 from:</i> • Chevron (serrated) nozzles mix bypass air with surrounding air more smoothly • Acoustic liners in the engine cowlings absorb noise • Hush kits fitted to older low-bypass engines • Fairings on undercarriage and wing-flap hinges reduce turbulence; welded rather than riveted panels • Lighter composite aircraft need less thrust • Blended wing with engines on top so the body shields the ground	2
07.3	• Constant (3°) descent avoids periods of high thrust in a stepped descent and keeps aircraft higher for longer • Night flights (especially take-offs, which are noisier) are restricted / banned when people are sleeping and background noise is low • Noisier aircraft pay higher charges / use up more quota points and fines for exceeding limits, so airlines are encouraged to buy quieter aircraft • More time to slow down after landing so wheel brakes can be used instead of noisy reverse thrusters	4
07.4	• Engines run at higher power / maximum thrust and the steep climb angle needs full power (although the aircraft climbs quickly out of earshot)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 288-292; AQA spec 3.4.3.2.13.	
08.1	• High mean wind velocities that are consistent in strength and direction • A long fetch (distance of open water) over which waves build up, eg Atlantic coasts of the UK	2
08.2	• Oscillating water column (eg Islay Limpet): waves force water up and down in a submerged chamber; the air forced in and out drives a turbine • Or point absorber (floating buoy moving relative to a fixed base turns a generator); surface attenuator (Pelamis - hinged sections push pistons pumping fluid over a turbine); overtopping device (Wave Dragon - waves fill a reservoir above sea level which drains through turbines); oscillating wave surge converter (Oyster)	2
08.3	<i>Any 2 from:</i> • Equipment must withstand storms and corrosion • Difficult to anchor off stormy coasts / in deep water • Expensive to transmit electricity from isolated coasts to consumers • Competition with established technologies	2

Question	Marking guidance	Marks
08.4	• Barrage: huge quantities of construction materials; reduces tidal range so intertidal mudflats used by wading birds are lost; changes sedimentation (erosion near turbines, deposition elsewhere), lower turbidity; pollutants concentrate in the lagoon; blocks migratory fish and shipping • Lagoon: similar but smaller scale; only part of the estuary enclosed so the most sensitive areas and main channel can be avoided • In-stream turbines: do not block the tidal flow so little impact on range or sediment; but noise affects cetaceans and many dispersed turbines could have a larger total impact; harness less energy	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 210-211, 214-217; AQA spec 3.3.4.1.	
09.1	• Small seeds, genetically uniform, high and rapid germination rate so results are quick and reliable • Wild species have different ranges of tolerance so cress results cannot be applied directly	2
09.2	• Preliminary study with widely spaced pH values to find where germination occurs • Final range from below the lowest pH giving germination to above the highest, with closer intervals	2
09.3	• Temperature (constant, natural germination temperature) • Light, water volume, number of seeds per dish, seed source • Growth medium (filter paper; soil would alter pH); nutrients not needed for germination • Continue until it is clear that ungerminated seeds will not germinate, so that a delay in germination is not mistaken for failure	4
09.4	• Surface water affects fresh mass and drying damages the plants; dry mass requires killing the plants so the same plants cannot be re-measured • Leaf area is a reasonable estimate of biomass that allows repeated measurement of the same plants over time	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 416-418; AQA spec 3.7.2.2.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Uses / benefits: nuclear electricity (low carbon, high energy density), medicine (X-rays, CT, radiotherapy, tracers, sterilising equipment), industry (thickness gauges, polymer cross-linking, testing turbines, oil exploration), agriculture (food irradiation, sterile male technique, mutation breeding), research tracers, smoke detectors, ship / submarine propulsion • Risks: free radical production, DNA damage, mutation, cancer; acute and chronic, somatic and gonadic effects; contamination vs exposure; activation products; long-lived wastes; accidents (Chernobyl, Fukushima); weapons fallout • Factors affecting risk: type of radiation (alpha, beta, gamma, neutrons - penetration, RBE / weighting factors), half-life, dose (Gy, Sv), route of exposure • Risk:benefit analysis and why it is difficult (latency, alternative causes, poor exposure data, risks and benefits to different people, offsets such as avoided surgery) • Principles: ALARA, BATNEEC; closed sources, remote handling, absorbers / shielding, distance (inverse square law), time, protective clothing, decontamination, potassium iodide • Monitoring: dosimeters, film badges, air and contamination monitors; environmental monitoring via CPA and CGM; waste categories and disposal (vitrification, encapsulation, Drigg) • Evaluation / judgement: benefits generally outweigh risks when controls are applied; public perception vs actual risk (background radon 50% of dose vs <1% from nuclear industry)	9
10.2	• Consumers of the electricity may live in a different country / region from the uranium miners or the communities near waste stores who bear the risks	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 295-304; AQA spec 3.4.3.2.14.	

Question	Marking guidance	Marks
11.1	To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society? Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Fossil fuel features that shaped society: energy density, storage, transport, high temperatures, liquid vehicle fuels, established infrastructure • Renewable features: intermittent / unpredictable (wind, solar) vs predictable (tidal) vs controllable (HEP, biofuel, geothermal); low energy density; location; low pollution but manufacturing impacts • Technologies and developments for each resource with examples; new technologies improving usability • Storage and grid management: pumped storage, batteries, hydrogen economy, P2G, V2G, thermal storage, peak shaving; the role of electricity as a secondary fuel • Economics and politics: development costs, interest, subsidies, external costs of fossil fuels, falling renewable costs • Energy conservation as part of the solution; sustainability principles - natural systems run on low-density renewable energy • Judgement: electricity largely yes; heat, aviation, shipping and high-temperature industry harder; transition period with nuclear / CCS Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25

Question	Marking guidance	Marks
11.2	Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Features of fertile soil: water, nutrients, air, organic matter, pH, biota, texture, structure, depth; slow rate of soil formation vs rapid erosion (dynamic equilibrium broken) • Causes of degradation: vegetation removal, ploughing vulnerable soils, overgrazing, reduced organic matter and biota, cultivating steep slopes, compaction, salinisation from irrigation, nutrient depletion, pollution • Types of erosion: rain splash, runoff, wind, slumping; consequences: reduced productivity, river and reservoir sedimentation, flooding, coastal sedimentation, particulates, desertification, landslides • USLE as a tool for estimating loss and testing management • Management: long-term crops, low / zero tillage, contour ploughing, tied ridging, terracing, stone rows, windbreaks, multicropping / strip cropping, mulching, organic matter, livestock control, drainage / pH / nutrient management, conservation of soil biota • Links to sustainable agriculture, carbon storage, hydrological cycle, circular economy • Evaluation and judgement on which strategies are most effective in different environments Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169, 328-337; AQA spec 3.2.5, 3.5.1.3, 3.5.1.5.	25